



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | rotarod            |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.15.0 |
| Date            | 07/20/2026         |

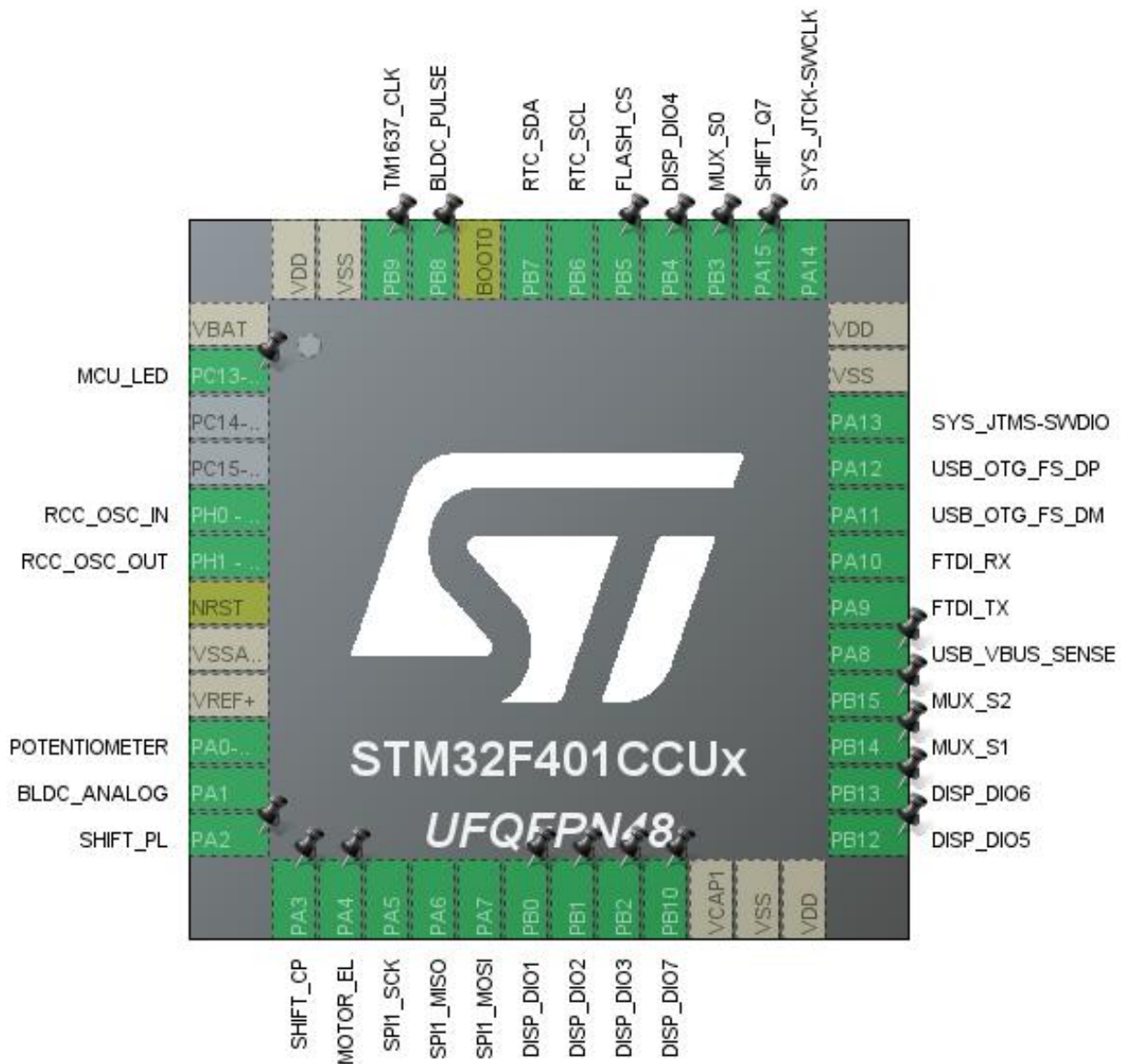
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F4       |
| MCU Line       | STM32F401     |
| MCU name       | STM32F401CCUx |
| MCU Package    | UFQFPN48      |
| MCU Pin number | 48            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration



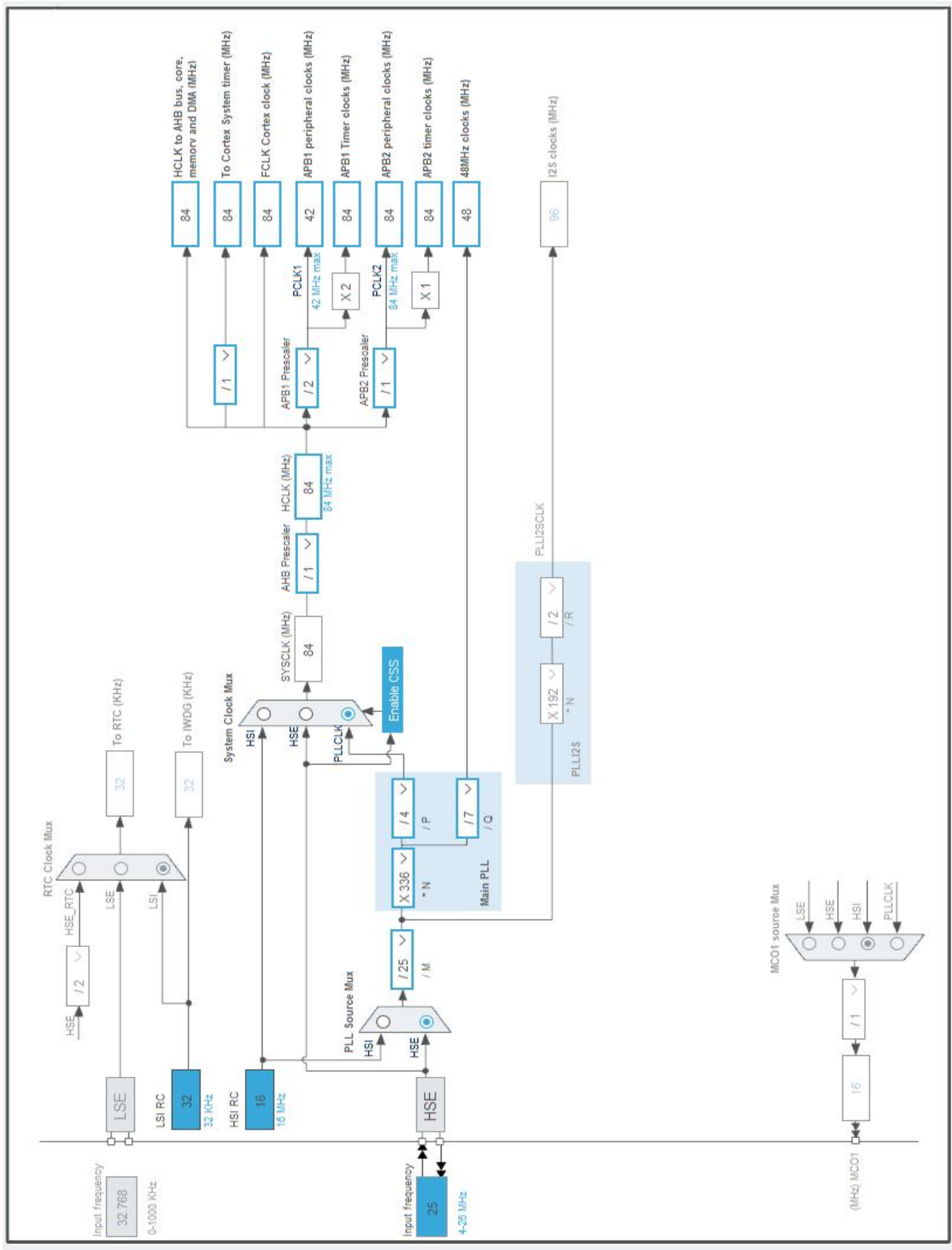
### 3. Pins Configuration

| Pin Number<br>UFQFPN48 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label          |
|------------------------|---------------------------------------|----------|--------------------------|----------------|
| 1                      | VBAT                                  | Power    |                          |                |
| 2                      | PC13-ANTI_TAMP *                      | I/O      | GPIO_Output              | MCU_LED        |
| 5                      | PH0 - OSC_IN                          | I/O      | RCC_OSC_IN               |                |
| 6                      | PH1 - OSC_OUT                         | I/O      | RCC_OSC_OUT              |                |
| 7                      | NRST                                  | Reset    |                          |                |
| 8                      | VSSA/VREF-                            | Power    |                          |                |
| 9                      | VREF+                                 | Power    |                          |                |
| 10                     | PA0-WKUP                              | I/O      | ADC1_IN0                 | POTENTIOMETER  |
| 11                     | PA1                                   | I/O      | TIM5_CH2                 | BLDC_ANALOG    |
| 12                     | PA2 *                                 | I/O      | GPIO_Output              | SHIFT_PL       |
| 13                     | PA3 *                                 | I/O      | GPIO_Output              | SHIFT_CP       |
| 14                     | PA4 *                                 | I/O      | GPIO_Output              | MOTOR_EL       |
| 15                     | PA5                                   | I/O      | SPI1_SCK                 |                |
| 16                     | PA6                                   | I/O      | SPI1_MISO                |                |
| 17                     | PA7                                   | I/O      | SPI1_MOSI                |                |
| 18                     | PB0 *                                 | I/O      | GPIO_Output              | DISP_DIO1      |
| 19                     | PB1 *                                 | I/O      | GPIO_Output              | DISP_DIO2      |
| 20                     | PB2 *                                 | I/O      | GPIO_Output              | DISP_DIO3      |
| 21                     | PB10 *                                | I/O      | GPIO_Output              | DISP_DIO7      |
| 22                     | VCAP1                                 | Power    |                          |                |
| 23                     | VSS                                   | Power    |                          |                |
| 24                     | VDD                                   | Power    |                          |                |
| 25                     | PB12 *                                | I/O      | GPIO_Output              | DISP_DIO5      |
| 26                     | PB13 *                                | I/O      | GPIO_Output              | DISP_DIO6      |
| 27                     | PB14 *                                | I/O      | GPIO_Output              | MUX_S1         |
| 28                     | PB15 *                                | I/O      | GPIO_Output              | MUX_S2         |
| 29                     | PA8 *                                 | I/O      | GPIO_Input               | USB_VBUS_SENSE |
| 30                     | PA9                                   | I/O      | USART1_TX                | FTDI_TX        |
| 31                     | PA10                                  | I/O      | USART1_RX                | FTDI_RX        |
| 32                     | PA11                                  | I/O      | USB_OTG_FS_DM            |                |
| 33                     | PA12                                  | I/O      | USB_OTG_FS_DP            |                |
| 34                     | PA13                                  | I/O      | SYS_JTMS-SWDIO           |                |
| 35                     | VSS                                   | Power    |                          |                |
| 36                     | VDD                                   | Power    |                          |                |
| 37                     | PA14                                  | I/O      | SYS_JTCK-SWCLK           |                |
| 38                     | PA15 *                                | I/O      | GPIO_Input               | SHIFT_Q7       |

| Pin Number<br>UFQFPN48 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label      |
|------------------------|---------------------------------------|----------|--------------------------|------------|
| 39                     | PB3 *                                 | I/O      | GPIO_Output              | MUX_S0     |
| 40                     | PB4 *                                 | I/O      | GPIO_Output              | DISP_DIO4  |
| 41                     | PB5 *                                 | I/O      | GPIO_Output              | FLASH_CS   |
| 42                     | PB6                                   | I/O      | I2C1_SCL                 | RTC_SCL    |
| 43                     | PB7                                   | I/O      | I2C1_SDA                 | RTC_SDA    |
| 44                     | BOOT0                                 | Boot     |                          |            |
| 45                     | PB8                                   | I/O      | TIM4_CH3                 | BLDC_PULSE |
| 46                     | PB9 *                                 | I/O      | GPIO_Output              | TM1637_CLK |
| 47                     | VSS                                   | Power    |                          |            |
| 48                     | VDD                                   | Power    |                          |            |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F4       |
| Line      | STM32F401     |
| MCU       | STM32F401CCUx |
| Datasheet | DS9716 Rev8   |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.3 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

#### 1.4. Sequence

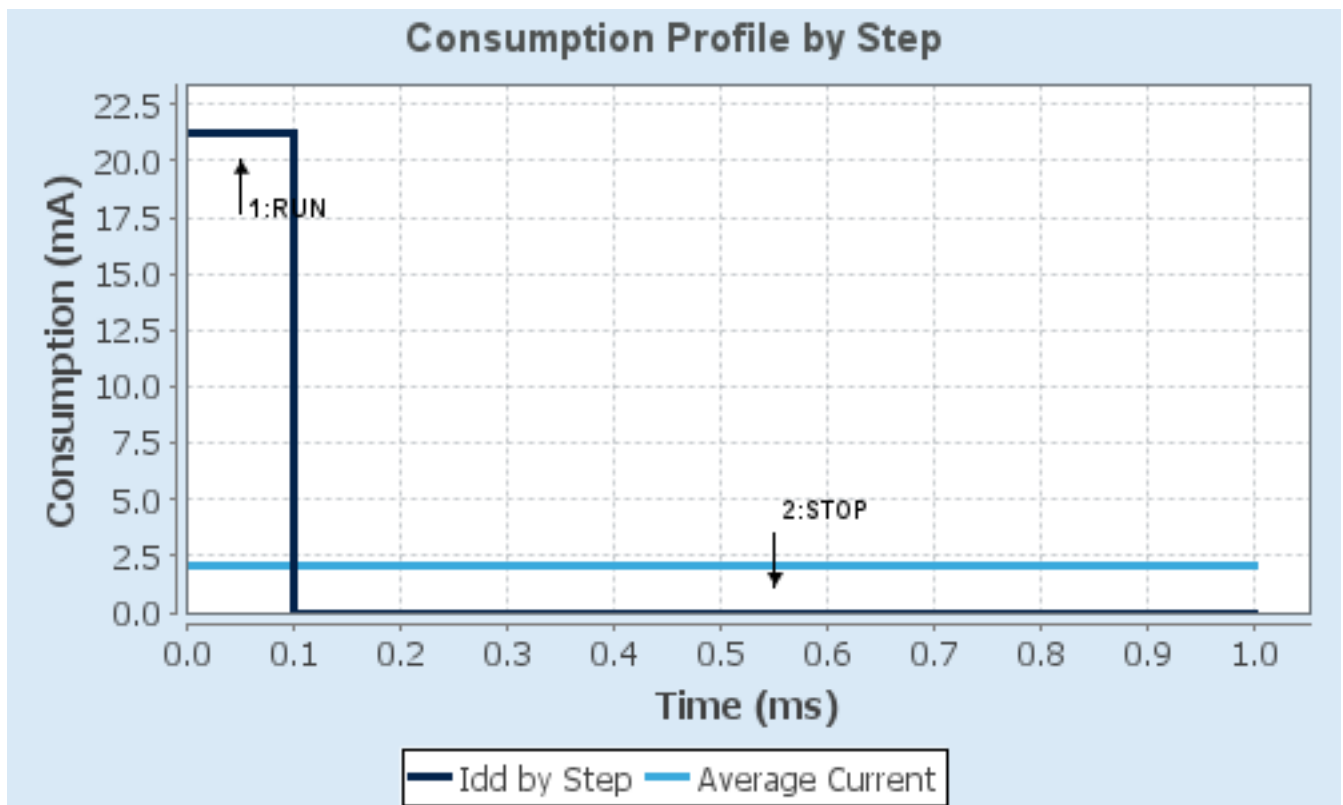
|                               |                    |                             |
|-------------------------------|--------------------|-----------------------------|
| <b>Step</b>                   | Step1              | Step2                       |
| <b>Mode</b>                   | RUN                | STOP                        |
| <b>Vdd</b>                    | 3.3                | 3.3                         |
| <b>Voltage Source</b>         | Battery            | Battery                     |
| <b>Range</b>                  | Scale2-Medium      | No Scale                    |
| <b>Fetch Type</b>             | FLASH/ART/PREFETCH | n/a                         |
| <b>CPU Frequency</b>          | 84 MHz             | 0 Hz                        |
| <b>Clock Configuration</b>    | HSE PLL            | Regulator_LPLV Flash-PwrDwn |
| <b>Clock Source Frequency</b> | 4 MHz              | 0 Hz                        |
| <b>Peripherals</b>            |                    |                             |
| <b>Additional Cons.</b>       | 0 mA               | 0 mA                        |
| <b>Average Current</b>        | 21.2 mA            | 10 $\mu$ A                  |
| <b>Duration</b>               | 0.1 ms             | 0.9 ms                      |
| <b>DMIPS</b>                  | 105.0              | 0.0                         |
| <b>Ta Max</b>                 | 102.76             | 105                         |
| <b>Category</b>               | In DS Table        | In DS Table                 |

#### 1.5. Results

|               |                               |                 |             |
|---------------|-------------------------------|-----------------|-------------|
| Sequence Time | 1 ms                          | Average Current | 2.13 mA     |
| Battery Life  | 2 months, 5 days,<br>14 hours | Average DMIPS   | 105.0 DMIPS |

#### 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                               |
|-----------------------------------|-----------------------------------------------------|
| Project Name                      | rotarod                                             |
| Project Folder                    | C:\Users\USER\STM32CubeIDE\workspace_1.19.0\rotarod |
| Toolchain / IDE                   | STM32CubeIDE                                        |
| Firmware Package Name and Version | STM32Cube FW_F4 V1.28.3                             |
| Application Structure             | Advanced                                            |
| Generate Under Root               | Yes                                                 |
| Do not generate the main()        | No                                                  |
| Minimum Heap Size                 | 0x200                                               |
| Minimum Stack Size                | 0x400                                               |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name          | Peripheral Instance Name |
|------|------------------------|--------------------------|
| 1    | SystemClock_Config     | RCC                      |
| 2    | MX_GPIO_Init           | GPIO                     |
| 3    | MX_USB_OTG_FS_PCD_Init | USB_OTG_FS               |
| 4    | MX_ADC1_Init           | ADC1                     |
| 5    | MX_I2C1_Init           | I2C1                     |
| 6    | MX_SPI1_Init           | SPI1                     |
| 7    | MX_TIM5_Init           | TIM5                     |
| 8    | MX_USART1_UART_Init    | USART1                   |
| 9    | MX_TIM4_Init           | TIM4                     |



## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

mode: IN0

#### 3.1.1. Parameter Settings:

##### ADCs\_Common\_Settings:

Mode Independent mode

##### ADC\_Settings:

Clock Prescaler PCLK2 divided by 4

Resolution 12 bits (15 ADC Clock cycles)

Data Alignment Right alignment

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

End Of Conversion Selection EOC flag at the end of single channel conversion

##### ADC\_Regular\_ConversionMode:

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 0

Sampling Time 3 Cycles

##### ADC\_Injected\_ConversionMode:

Number Of Conversions 0

##### WatchDog:

Enable Analog WatchDog Mode false

### 3.2. I2C1

I2C: I2C

#### 3.2.1. Parameter Settings:

##### Master Features:

I2C Speed Mode Standard Mode

I2C Clock Speed (Hz) 100000

##### Slave Features:

Clock No Stretch Mode Disabled

Primary Address Length selection 7-bit

|                                |          |
|--------------------------------|----------|
| Dual Address Acknowledged      | Disabled |
| Primary slave address          | 0        |
| General Call address detection | Disabled |

### 3.3. RCC

#### High Speed Clock (HSE): Crystal/Ceramic Resonator

##### 3.3.1. Parameter Settings:

###### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Enabled            |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 2 WS (3 CPU cycle) |

###### RCC Parameters:

|                                |          |
|--------------------------------|----------|
| HSI Calibration Value          | 16       |
| TIM Prescaler Selection        | Disabled |
| HSE Startup Timeout Value (ms) | 100      |
| LSE Startup Timeout Value (ms) | 5000     |

###### Power Parameters:

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 2 |
|-------------------------------|---------------------------------|

### 3.4. SPI1

#### Mode: Full-Duplex Master

##### 3.4.1. Parameter Settings:

###### Basic Parameters:

|              |           |
|--------------|-----------|
| Frame Format | Motorola  |
| Data Size    | 8 Bits    |
| First Bit    | MSB First |

###### Clock Parameters:

|                           |                       |
|---------------------------|-----------------------|
| Prescaler (for Baud Rate) | 2                     |
| Baud Rate                 | <b>42.0 MBits/s *</b> |
| Clock Polarity (CPOL)     | Low                   |
| Clock Phase (CPHA)        | 1 Edge                |

###### Advanced Parameters:

|                 |          |
|-----------------|----------|
| CRC Calculation | Disabled |
|-----------------|----------|

NSS Signal Type

Software

### 3.5. SYS

**Debug: Serial Wire**

**Timebase Source: SysTick**

### 3.6. TIM4

**Channel3: Input Capture direct mode**

#### 3.6.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |               |
|-------------------------------------------------------|---------------|
| Prescaler (PSC - 16 bits value)                       | <b>8399 *</b> |
| Counter Mode                                          | Up            |
| Counter Period (AutoReload Register - 16 bits value ) | 65535         |
| Internal Clock Division (CKD)                         | No Division   |
| auto-reload preload                                   | Disable       |

##### **Trigger Output (TRGO) Parameters:**

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

##### **Input Capture Channel 3:**

|                             |             |
|-----------------------------|-------------|
| Polarity Selection          | Rising Edge |
| IC Selection                | Direct      |
| Prescaler Division Ratio    | No division |
| Input Filter (4 bits value) | 0           |

### 3.7. TIM5

**Channel2: PWM Generation CH2**

#### 3.7.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |                 |
|-------------------------------------------------------|-----------------|
| Prescaler (PSC - 16 bits value)                       | 0               |
| Counter Mode                                          | Up              |
| Counter Period (AutoReload Register - 32 bits value ) | <b>4199 *</b>   |
| Internal Clock Division (CKD)                         | No Division     |
| auto-reload preload                                   | <b>Enable *</b> |

##### **Trigger Output (TRGO) Parameters:**

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

#### **PWM Generation Channel 2:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (32 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

### **3.8. USART1**

#### **Mode: Asynchronous**

##### 3.8.1. Parameter Settings:

#### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

#### **Advanced Parameters:**

|                |                      |
|----------------|----------------------|
| Data Direction | Receive and Transmit |
| Over Sampling  | 16 Samples           |

### **3.9. USB\_OTG\_FS**

#### **Mode: Device\_Only**

##### 3.9.1. Parameter Settings:

|                       |                            |
|-----------------------|----------------------------|
| Speed                 | Device Full Speed 12MBit/s |
| Low power             | Disabled                   |
| Link Power Management | Disabled                   |
| VBUS sensing          | Disabled                   |
| Signal start of frame | Disabled                   |

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP         | Pin            | Signal         | GPIO mode                     | GPIO pull/up pull down      | Max Speed   | User Label    |
|------------|----------------|----------------|-------------------------------|-----------------------------|-------------|---------------|
| ADC1       | PA0-WKUP       | ADC1_IN0       | Analog mode                   | No pull-up and no pull-down | n/a         | POTENTIOMETER |
| I2C1       | PB6            | I2C1_SCL       | Alternate Function Open Drain | No pull-up and no pull-down | Very High * | RTC_SCL       |
|            | PB7            | I2C1_SDA       | Alternate Function Open Drain | No pull-up and no pull-down | Very High * | RTC_SDA       |
| RCC        | PH0 - OSC_IN   | RCC_OSC_IN     | n/a                           | n/a                         | n/a         |               |
|            | PH1 - OSC_OUT  | RCC_OSC_OUT    | n/a                           | n/a                         | n/a         |               |
| SPI1       | PA5            | SPI1_SCK       | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * |               |
|            | PA6            | SPI1_MISO      | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * |               |
|            | PA7            | SPI1_MOSI      | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * |               |
| SYS        | PA13           | SYS_JTMS-SWDIO | n/a                           | n/a                         | n/a         |               |
|            | PA14           | SYS_JTCK-SWCLK | n/a                           | n/a                         | n/a         |               |
| TIM4       | PB8            | TIM4_CH3       | Alternate Function Push Pull  | No pull-up and no pull-down | Low         | BLDC_PULSE    |
| TIM5       | PA1            | TIM5_CH2       | Alternate Function Push Pull  | No pull-up and no pull-down | Low         | BLDC_ANALOG   |
| USART1     | PA9            | USART1_TX      | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * | FTDI_TX       |
|            | PA10           | USART1_RX      | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * | FTDI_RX       |
| USB_OTG_FS | PA11           | USB_OTG_FS_DM  | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * |               |
|            | PA12           | USB_OTG_FS_DP  | Alternate Function Push Pull  | No pull-up and no pull-down | Very High * |               |
| GPIO       | PC13-ANTI_TAMP | GPIO_Output    | Output Push Pull              | No pull-up and no pull-down | Low         | MCU_LED       |
|            | PA2            | GPIO_Output    | Output Push Pull              | No pull-up and no pull-down | Low         | SHIFT_PL      |
|            | PA3            | GPIO_Output    | Output Push Pull              | No pull-up and no pull-down | Low         | SHIFT_CP      |
|            | PA4            | GPIO_Output    | Output Push Pull              | No pull-up and no pull-down | Low         | MOTOR_EL      |
|            | PB0            | GPIO_Output    | Output Push Pull              | No pull-up and no pull-down | Low         | DISP_DIO1     |
|            |                |                |                               |                             |             |               |



| IP | Pin  | Signal      | GPIO mode        | GPIO pull/up pull down      | Max Speed | User Label     |
|----|------|-------------|------------------|-----------------------------|-----------|----------------|
|    | PB1  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | DISP_DIO2      |
|    | PB2  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | DISP_DIO3      |
|    | PB10 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | DISP_DIO7      |
|    | PB12 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | DISP_DIO5      |
|    | PB13 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | DISP_DIO6      |
|    | PB14 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | MUX_S1         |
|    | PB15 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | MUX_S2         |
|    | PA8  | GPIO_Input  | Input mode       | No pull-up and no pull-down | n/a       | USB_VBUS_SENSE |
|    | PA15 | GPIO_Input  | Input mode       | No pull-up and no pull-down | n/a       | SHIFT_Q7       |
|    | PB3  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | MUX_S0         |
|    | PB4  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | DISP_DIO4      |
|    | PB5  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | FLASH_CS       |
|    | PB9  | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | TM1637_CLK     |

## 4.2. DMA configuration

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault    | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 15                   | 0           |
| TIM4 global interrupt                   | true   | 0                    | 0           |
| PVD interrupt through EXTI line 16      | unused |                      |             |
| Flash global interrupt                  | unused |                      |             |
| RCC global interrupt                    | unused |                      |             |
| ADC1 global interrupt                   | unused |                      |             |
| I2C1 event interrupt                    | unused |                      |             |
| I2C1 error interrupt                    | unused |                      |             |
| SPI1 global interrupt                   | unused |                      |             |
| USART1 global interrupt                 | unused |                      |             |
| TIM5 global interrupt                   | unused |                      |             |
| USB On The Go FS global interrupt       | unused |                      |             |
| FPU global interrupt                    | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Pre-fetch fault, memory access fault    | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |
| TIM4 global interrupt                   | false                             | true                 | true             |

**\* User modified value**

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

#### Middleware

#### System Core

#### Analog

#### Timers

#### Connectivity

#### Multimedia

#### Computing

DMA

ADC1 

TIM4 

I2C1 

GPIO 

TIM5 

SP1 

IIVIC 

USART1 

RCC 

USB\_FS 

SYS 

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32f401_bsdl.zip">https://www.st.com/resource/en/bsdl_model/stm32f401_bsdl.zip</a>                                                                                                                                                                                                 |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32f401_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32f401_ibis.zip</a>                                                                                                                                                                                                 |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32f4-svd.zip">https://www.st.com/resource/en/svd/stm32f4-svd.zip</a>                                                                                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                                             |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                                             |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32f4x1.pdf">https://www.st.com/resource/en/flyer/flstm32f4x1.pdf</a>                                                                                                                                                                                                                 |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                             |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                                                                               |
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